

Problem Set 1

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Due: October 8, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 8, 2026. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,
      80, 97, 95, 111, 114, 89, 95, 126, 98)
```

1. Find a 90% confidence interval for the average student IQ in the school (As a hint, you first need the mean and SD to create a CI):

```
1 #Q1, 1. Construct a 90% confidence interval for the mean IQ.
2 # CI = sample mean +/- critical value * SE
3
4 #step 1: For CI, must calculate the sample mean of IQs
5 y_samp_mean <- mean(y)
```

```

6 y_samp_mean #sample mean is 98.44
7
8 #step 2: then find the sample sd using the sample mean
9 y_sd <- sd(y)
10 y_sd #sd of y is 13.09287
11
12 #step 3: get n (sample size of students)
13 n <- length(y)
14 n #n = 25 -> a small sample
15
16 #step 4: find standard error (estimates the SD of the sampling
      distribution)
17 y_se <- y_sd / sqrt(n)
18 y_se #standard error = 2.618575
19
20 #step 5: find the critical t-value
21 # population SD is unknown (we estimate it from the sample) and n is
      small,
22 # so using the t distribution with df = n - 1 = 24
23 # for a 90% CI I leave 5% in each tail, so need the value at p = 0.95
24 alpha <- 0.10
25 t_val <- qt(1 - alpha/2, df = n-1)
26 t_val #t value = 1.710882
27
28 #step 6: margin of error = critical t value * SE
29 margin_error <- t_val * y_se
30 margin_error #margin_error = 4.480072
31
32 #step 7: use margin of error to find the lower and higher ends of CI
33 lower_end_CI <- y_samp_mean - margin_error
34 upper_end_CI <- y_samp_mean + margin_error
35 lower_end_CI #lower end CI is 93.95993
36 upper_end_CI #upper end CI is 102.9201
37 #CI is [93.96, 102.92]
38
39 #check with the built-in function
40 t.test(y, conf.level = 0.90)
41 # ANSWER: 90% CI = [93.96, 102.92], same as by hand

```

The sample mean is 98.44, the sample standard deviation is 13.09, and the sample size is 25. The confidence interval is calculated as the sample mean $\pm$ the critical value $\times$ the standard error. Because the population standard deviation is unknown and has to be estimated from the sample (and the sample is small), I use the t-distribution with $df=n-1=24$ instead of the normal distribution. The t-distribution has thicker tails, so it gives a slightly wider interval, which accounts for the extra uncertainty from estimating the SD.

For a 90% confidence interval, $\alpha=0.10$, so I leave 5% in each tail and the t value is 1.71. The standard error is 2.619 ($13.09/\sqrt{25}$), and the margin of error is $1.71 \times 2.619 = 4.48$. Therefore, the confidence interval is 98.44 ± 4.48 , or 93.96 to 102.92. Using

the `t.test()` function gives the same interval.

So, I am 90% confident that the mean IQ of students at this school lies between 93.96 and 102.92. This means that if we took many random samples of 25 students and built an interval this way each time, about 90% of those intervals would contain the true mean IQ.

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

```
1 #Q1, 2. Test if school's mean IQ is greater than 100 (alpha=0.05)
2 # Hypotheses: one-sided, because the counselor asks whether the mean is
3 # higher than 100 (a specific direction), not just "different" from 100
4 # H0: average student IQ in the school is 100 (mu = 100)
5 # HA: average student IQ in the school is greater than 100 (mu > 100)
6
7 #step 1: use the t statistic = (sample mean - hypothesized mean) / SE
8 t_stat <- (y_samp_mean - 100) / y_se
9 t_stat #t_stat = -0.5957439
10
11 #step 2: find the p-val from the upper tail, since HA is mu > 100
12 # lower.tail = FALSE gives the probability of a t value at least this
    large
13 p_val <- pt(t_stat, df = n-1, lower.tail = FALSE)
14 p_val # p_val = 0.7215383
15
16 #check with the built-in function
17 t.test(y, mu = 100, alternative = "greater")
18
19 # ANSWER: p = 0.722 > 0.05 -> fail to reject H0. Not enough evidence that
    the
20 # school's mean IQ is greater than 100 (& the sample mean, 98.44, is
    below 100).
```

This is a one-sided t-test because the counselor wants to know whether the average student IQ in her school is *higher* than 100, which is a specific direction, not just whether it is different from 100. The null hypothesis is that average student IQ in her school = 100 ($H_0 : \mu = 100$), and the alternative hypothesis is that it's greater than 100 ($H_A : \mu > 100$). I use a t-test for the same reason as above: the population SD is unknown and estimated from the sample.

The test statistic is taken by comparing the sample mean to 100 and then dividing by standard error: $(98.44 - 100)/2.619 = -0.596$. Because H_A is $\mu > 100$, the p-value is the area in the upper tail of the t-distribution (when `lower.tail = FALSE`). Using 24 degrees of freedom, the one-sided p-value is 0.722.

Since $p=0.722$ is greater than $\alpha=0.05$, I fail to reject the null hypothesis. There is insufficient evidence at the 5% significance level to conclude that the mean IQ of

students at this school is greater than 100. In addition, the sample mean (98.44) is below 100, so the sample doesn't point in the direction of the alternative hypothesis at all.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

State	50 states in US
Y	per capita expenditure on shelters/housing assistance in state
X1	per capita personal income in state
X2	Number of residents per 100,000 that are "financially insecure" in state
X3	Number of people per thousand residing in urban areas in state
Region	1=Northeast, 2= North Central, 3= South, 4=West

Explore the `expenditure` data set and import data into R.

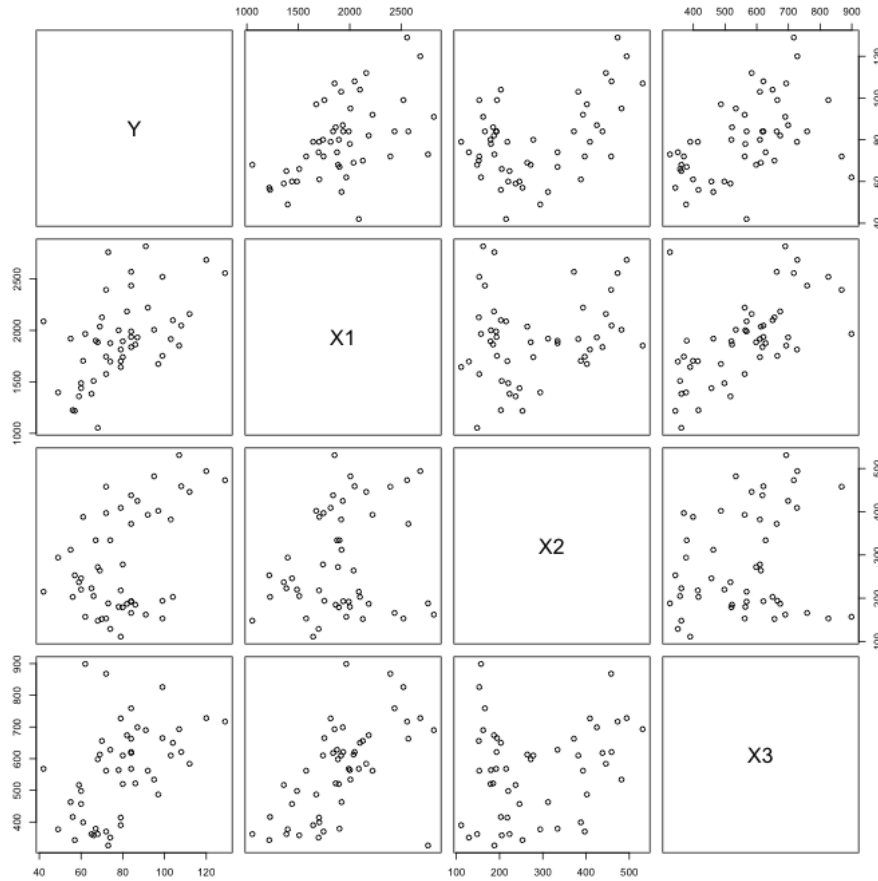
```
1 expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2026/main/datasets/expenditure.txt", header=T)
```

```
1 #explore data
2 head(expenditure)
3 summary(expenditure)
4 str(expenditure)
```

- Please plot the relationships among Y , $X1$, $X2$, and $X3$? What are the correlations among them (you just need to describe the graph and the relationships among them)?

```
1 #Q2, 1. Plot relationships among vars. What are the correlations?
2 #step 1: subset the dataset to only look at the variables I want
3 # plot() on a df of numeric columns makes a scatterplot matrix
4
5 png("graph_1.png", width = 700, height = 700)
6 expenditure %>%
7   select(Y, X1, X2, X3) %>%
8   plot()
9 dev.off()
10
11 #step 2: calculate the correlations among them
12 expenditure %>%
13   select(Y, X1, X2, X3) %>%
14   cor()
15 # All positive. Y-X1 = 0.532, Y-X2 = 0.448, Y-X3 = 0.464,
16 # X1-X2 = 0.206, X1-X3 = 0.595 (strongest), X2-X3 = 0.221
```

Figure 1: Scatterplot matrix of Y, X1, X2, and X3.



The scatterplot matrix in plot 1 shows positive relationships between all four variables, and the correlations calculated with `cor()` confirm this. Y (per capita expenditure on housing assistance) has moderate positive relationships with X1 (per capita personal income) (0.532), X2 (the number of financially insecure residents per 100,000) (0.448), and X3 (the number of people per 1,000 in urban areas) (0.464). This means that higher values of each of these variables generally correspond with higher housing expenditures, although there is substantial variation. The strongest relationship is between X1 and X3, with a correlation of 0.595, suggesting that states with higher per capita personal income also tend to have a larger share of their population living in urban areas. The relationships between X1 and X2 (0.206) and between X2 and X3 (0.221) are comparatively weak. Overall, all of the relationships are positive, but none are especially strong, and the graphs show considerable spread around these general patterns.

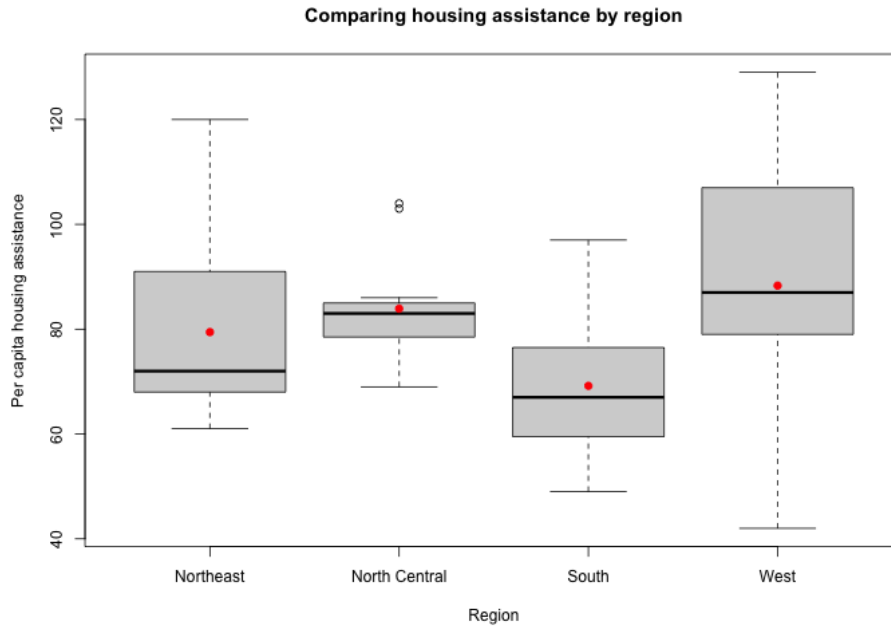
- Please plot the relationship between *Y* and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

```

1 #Q2, 2. Plot relationship btwn Y and region. Which region has highest p/c
2 #expenditure on housing assistance
3
4 #step 1: make region (stored as numeric) a factor
5 class(expenditure$Region)
6 expenditure$Region <- factor(
7   expenditure$Region,
8   levels = c(1, 2, 3, 4),
9   labels = c("Northeast", "North Central", "South", "West")
10 )
11
12 #step 2: y is numeric, region is categorical —> use boxplot
13 png("graph_y_region.png", width = 700, height = 500)
14 plot(expenditure$Region, expenditure$Y,
15       main = "Comparing housing assistance by region",
16       xlab = "Region",
17       ylab = "Per capita housing assistance")
18 points(tapply(expenditure$Y, expenditure$Region, mean), col = "red", pch
19         = 19)
19 dev.off()
20
21 #The red dots show the mean of each region and the black lines show the
22   median.
23 #The West has the highest median and mean (about 88.3) but also the
24   widest
25 #spread, while the South has the lowest median and mean (about 69.2).
26   North
27 #Central's values are tightly clustered, with two high outliers. The
28   Northeast's
29 # mean (79.4) is well above its median, so a few high-spending states are
30 #pulling its average up.
31
32 #Step 3: confirm means by region
33 expenditure %>%
34   group_by(Region) %>%
35   summarise(mean_y = mean(Y))
36 #mean findings: NE = 79.4, NC = 83.9, S = 69.2, W = 88.3
37 # The West has the highest average per capita expenditure on housing
38   assistance

```

Figure 2: Boxplot of Y by Region (red dots show the mean of each region).



I use a boxplot because Y is numeric and Region is categorical. Plot 2 shows differences in housing assistance expenditure across the four regions, although there is also variation within each region. The South generally has lower expenditures, while the West and North Central regions tend to have higher expenditures. The West also has the widest spread, while North Central's values are tightly clustered, with two high outliers.

Since the boxplot shows the median of each region rather than the mean, I also calculated the mean for each region and added it to the boxplot as a red dot. The mean per capita expenditure on housing assistance is approximately 79.4 in the Northeast, 83.9 in North Central, 69.2 in the South, and 88.3 in the West. Therefore, the West has the highest average per capita expenditure on housing assistance, with a mean of approximately 88.3. The Northeast's median is noticeably lower than its mean, which may suggest that a few states with high expenditure are pulling its average up.

- Please plot the relationship between Y and X1? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

```

1 #Q2,3. Plot relationship btwn Y and X1 Describe graph + relationship.
  Then,
2 #reproduce with Region and display w/ diff symbols and colors.
3
4 #Step 1: y is numerical, X1 is numerical -> use scatterplot
5 png("graph_y_X1.png", width = 700, height = 500)

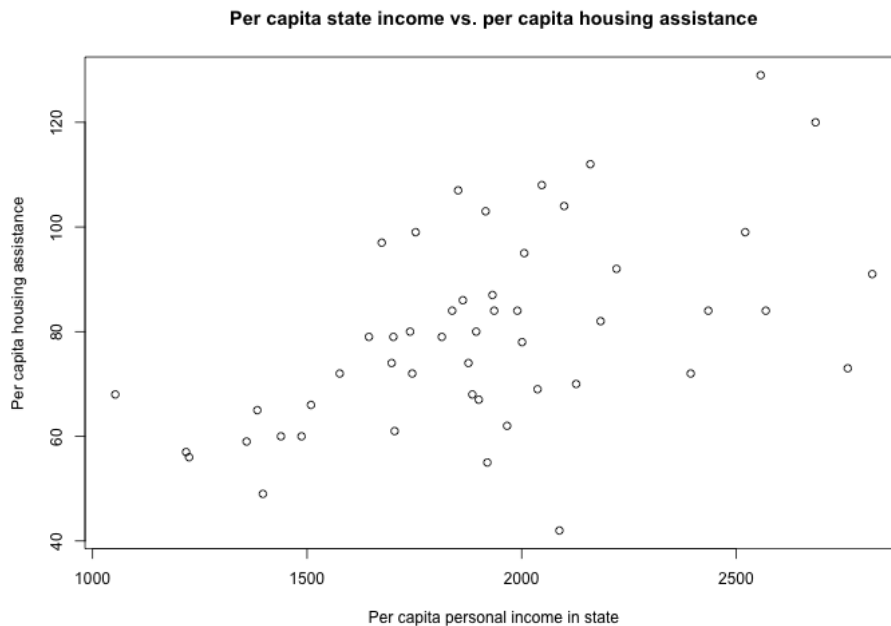
```

```

6 plot(expenditure$X1, expenditure$Y,
7       main = "Per capita state income vs. per capita housing assistance",
8       xlab = "Per capita personal income in state",
9       ylab = "Per capita housing assistance")
10 dev.off()
11 #The scatterplot shows a moderate positive relationship (r = 0.532), but
    points
12 #are spread out

```

Figure 3: Scatterplot of Y and X1.



I use a scatterplot because both variables are numeric. Plot 3 shows a moderate positive relationship between per capita personal income (X1) and per capita expenditure on housing assistance (Y), with a correlation of 0.532. States with higher personal income generally tend to have higher housing assistance expenditures, although the points are fairly spread out, so the relationship is not especially strong.

```

1 #Step 2: add region with different symbols
2
3 png("graph_y_X1_region.png", width = 700, height = 500)
4 plot(expenditure$X1, expenditure$Y,
5       col = as.numeric(expenditure$Region),
6       pch = as.numeric(expenditure$Region),
7       main = "Housing assistance and income by region",
8       xlab = "Per capita personal income in state",
9       ylab = "Per capita housing assistance")
10 legend("topleft",
11       legend = levels(expenditure$Region),

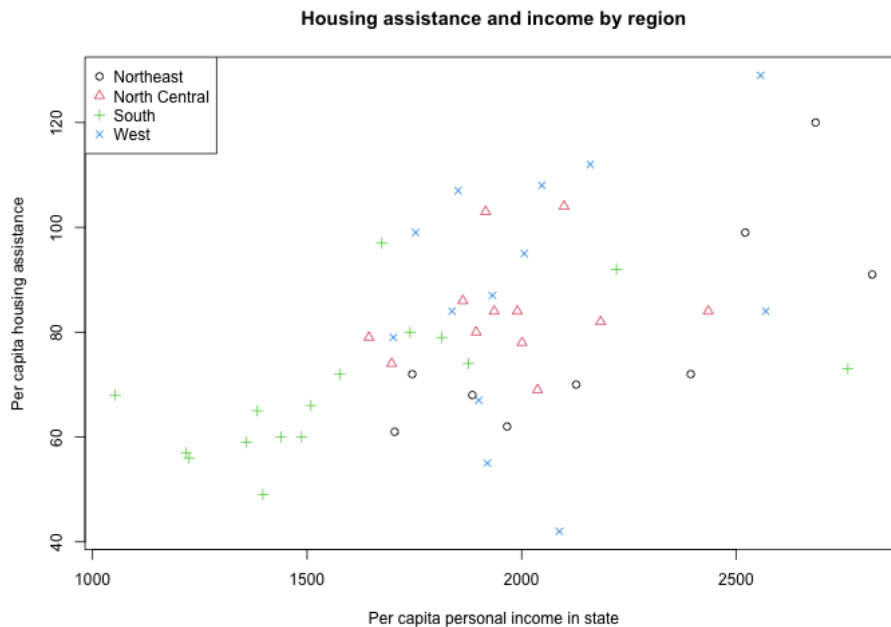
```

```

12     col = 1:4,
13     pch = 1:4)
14 dev.off()
15 #as.numeric() turns the 4 regions into 1–4, which plot() uses as the
    color and
16 #symbol (pch) codes, so each region gets its own color and symbol
17 #legend() added because plot() doesn't make one automatically
18
19 #The South's states cluster at low income and low spending, and show the
20 #clearest positive trend. North Central's states are in the middle on
    both.
21 #The Northeast mostly has higher incomes, but its spending varies a lot.
22 #The West is the most spread out, with spending from about 42 to 129 at
23 #similar income levels, so the relationship between X1 and Y looks
24 #different depending on the region.

```

Figure 4: Scatterplot of Y and X1, with regions shown by different colors and symbols.



Plot 4 reproduces the same scatterplot, with each region displayed using a different color and symbol. The South's states cluster at low income and low spending, and show the clearest positive trend. North Central's states are in the middle on both. The Northeast mostly has higher incomes, but its spending varies a lot. The West is the most spread out, with spending from about 42 to 129 at similar income levels. This suggests the relationship between X1 and Y looks different depending on the region.